

Product Name:

DOXYVET 50% Water Soluble powder

Composition:

Each gram contains:

Doxycycline 500mg

Product Description:

Yellow powder

Pharmacodynamics:

Doxycycline is a bacteriostatic antibiotic that acts by interfering with the bacterial protein synthesis of sensitive species.

Doxycycline is a semi-synthetic tetracycline derived from oxytetracycline. It acts on the subunit 30 S of the bacterial ribosome, to which is bound reversibly, blocking the union between aminoacyl-tRNA(transfer RNA) to the mRNA-ribosome complex, preventing the addition of new aminoacids into the growing peptide chain and thus interfering with protein synthesis.

Pharmacokinetics:

Doxycycline is bio-available after oral administration. When orally administered, it reaches values greater than 70% in most species.

Feeding can modify the oral bioavailability of Doxycycline. In fasting conditions bioavailability is around 10-15% greater than when the animal is fed. Doxycycline is well distributed through the body as it is highly lipid soluble. It reaches well irrigated tissues as well as peripheral ones. It accumulates in liver, kidney, bones and intestine; enterohepatic recycling occurs. In lungs it always reaches higher concentrations than in plasma. Therapeutic concentrations have been detected in aqueous humour, myocardium, reproductive tissues, brain and mammary gland. Plasma protein binding is 90-92%. 40% of drug is metabolized and largely excreted through faeces (biliary and intestinal route), mainly as microbiologically inactive conjugates.

Indications:

Chickens (broilers) : Treatment of Chronic Respiratory Disease (CRD) caused by Mycoplasma gallisepticum. Fattening pigs: Treatment of clinical respiratory infection caused by sensitive strains of Pasteurella multocida.

Dosage and Administration:

In drinking water use.

Chickens (broilers): 20 mg of doxycycline (equivalent to 40 mg of the veterinary medicinal product) / kg BW / day for 3 - 5 days

Fattening pigs: 10 mg of doxycycline (equivalent to 20 mg of the veterinary medicinal product) / kg b.w / day for 5 days.

For the preparation of the medicated water the body weight of the animals to be treated and their actual daily water intake should be taken into due account. Consumption may vary depending on factors like age, state of health, breed, husbandry system. To provide the required amount of active substance in mg per litre drinking water the following calculation should be made:

$$\frac{\text{mg doxycycline / kg bodyweight / day} \times \text{Average body weight (kg) of the animals to be treated}}{\text{Average amount of drinking water / animal}} = \text{Mg doxycycline 1 of drinking water}$$

To ensure a correct dosage body weight should be determined as accurately as possible to avoid underdosing. The intake of medicated water depends on the clinical condition of the animals. In order to obtain the correct dosage the concentration of doxycycline has to be adjusted accordingly. Sufficient access to the system of water supply should be available for the animals to be treated to ensure adequate water consumption. No other source of drinking water should be available during the medication period. Only sufficient medicated drinking water should be prepared to cover daily requirements

Contraindications:

Do not use in animals with known hypersensitivity to tetracyclines.
Do not use in animals with hepatic disorders.
Do not use in animals with renal disorders

Interaction with Other Medicaments:

Do not administer together with bactericidal antibiotics (penicillins, aminoglycosides, etc.)
Absorption of doxycycline can be decreased in the presence of high quantities of calcium, iron, magnesium or aluminium in the diet. Do not administered together with antacids, kaolin and iron preparations.
It is advised that the interval between the administration of other products containing polyvalent cations should be 1-2 hours because they limit the absorption of tetracyclines.
Doxycycline increases the action of anticoagulants.
The solubility of the product is pH dependent and will precipitate if mixed in alkaline solution.
Do not store the drinking water in metallic containers.

Pregnancy and Lactation:

Laboratory studies in rats and rabbits have not produced any evidence of a teratogenic, foetotoxic, maternotoxic effects.
The safety of the product has not been established in pregnant or lactating sows.
The product should not be used during pregnancy or lactation.
Do not use in laying birds producing eggs for human consumption

Side effects:

Allergic reactions.
Photosensitivity reactions.
Intestinal flora may be affected if treatment is very prolonged. This may result in digestive disorders.
If suspected adverse reactions occur, treatment should be discontinued. Inform your veterinary surgeon if adverse reactions occur that are not indicated.

Symptoms and Treatment of Overdose:

The administration of 40 mg/kg BW in pigs and 80 mg/kg in chickens (in both species corresponding to 4 times the recommended dose), for 5 days did not cause any adverse reaction.
In case of overdose the treatment should be suspended and symptomatic treatment established

Warning and Precautions:

Keep out of reach of children.

Jauhi dari kanak-kanak.

Special warnings (for each target species)

Due to likely variability (time, geographical) in susceptibility of bacteria to doxycycline, bacteriological sampling and susceptibility testing are recommended.

Under-dosing and/or treating for an insufficient length of time are considered to promote the development of resistance in bacteria and should be avoided.

Special precautions for use in animals

Avoid administration in oxidised drinking equipment

Use of the product should be based on susceptibility testing of the bacteria isolated from the animal. If this is not possible, therapy should be based on local (regional and farm level) epidemiological information about susceptibility of the target bacteria as well as by taking into account official national antimicrobial policies.

Inappropriate use of the product may increase the prevalence of bacteria resistant to doxycycline and may decrease the effectiveness of treatment with other tetracyclines due to the potential for cross-resistance.

Do not use at concentrations lower than 0.23 g of powder /l in drinking water with pH higher or equal to 7.5 to avoid precipitation.

Do not add acid to the medicated drinking water.

Withdrawal Periods:

Meat and offal

Pigs: 6 days

Chickens: 6 days

Not authorized for use in laying birds producing eggs for human consumption.

Do not use within 4 weeks of onset of the laying period.

Storage conditions:

Store in cool, dark place. Store below 30° C. Protect from direct light.

Pack Size:

1kg, 100gm

Distributed by Product Registration Holder:

LIM SENG PHARMACHEM SDN. BHD.

2692, Tingkat Perusahaan Enam,

Prai Industrial Estate MK1,

13600 Prai, Pulau Pinang, Malaysia.

Manufactured By:

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5, Lebuhr Perusahaan Klebang II,

Taman Perindustrian Antarabangsa IGB

31200 Chemor, Perak.

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